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1 Introduction

Fault-injection schedulers determine which behaviors a test can observe. A scheduler may choose one communication partition for a protocol view, one directed admission bit per peer pair, or distinct decisions for different voting phases. Twins, randomized Byzantine testing, Netrix, and ByzzBench demonstrate the practical value of controlled execution [2, 11, 6, 1]. Their capabilities also make a semantic question unavoidable: when does aggregating admission decisions preserve a safety counterexample?

Consider an enabling prevote and a later precommit carrying repair evidence. They share a directed peer connection and a protocol round. A phase-sensitive policy can admit the former and suppress the latter. An admission bit fixed across both phases admits both or neither. Admitting more evidence may change a validator's state before it issues a vote needed by a conflicting quorum. Thus counting possible schedules does not establish preservation of the observations that matter to a safety test.

The critical boundary is **vote issuance**. If a repair message arrives after an indispensable conflicting precommit has been signed, it cannot remove that precommit from another validator's future certificate. Repair before the second *decision* can therefore be too late. We call the collection of second-vote opportunities a **repair frontier** and study which pivotal contributions can jointly survive across it.

The one-pivot example is useful but does not characterize broader scheduler domains. With several initially ready pivots, a coarse policy can assign different validators to the two branches. With individual vote deadlines, a repair may be movable on its own but



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incompatible with delaying another repair. We connect these two effects. A graph derived from a finite opportunity partial order describes the compatible sets of unrepaired contributors; quorum deficits determine how large such a set must be.

1.1 Contributions and scope

The main contribution is an exact scheduler characterization for an explicit independent-pivot certificate schema. If there are k unit-weight pivots, the two quorums need d_0, d_1 pivotal contributions, and $\alpha(H)$ is the largest jointly reusable pivot set in the derived repair graph, a coarse unsafe execution exists exactly when

$$\max(d_0, d_1) \le k, \qquad d_0 + d_1 \le k + \alpha(H).$$

This yields the exact minimum number of independently phase-split channels required to recover a witness, a weighted extension, and a precise account of how relaxing delivery order changes the result. The graph reversal argument and its underlying optimization are classical; the contribution is their connection to admission coupling, persistent signed outputs, and quorum requirements.

We also repair the one-pivot criterion by stating persistence, enabling necessity, observation coverage, and the correct repair deadline explicitly. A complete four-validator execution realizes the mechanism using guarded Tendermint-style transitions under one omitted round-0 lock update. It includes actual signed vote sets, nil votes, legal timeout transitions, and a late-repair control whose two conflicting decisions occur at unmutated validators. Exhaustive finite checks and a scoped Lean development accompany these results.

The claims concern finite agreement observations and declared scheduling domains. The four-validator model is derived from pinned CometBFT rules; it does not execute the production client. The multi-pivot graph theorem assumes independent behavioral contracts and complete coverage of the stated opportunity orders. No universal ordering of test frameworks, unbounded completeness, liveness result, or deployed-client vulnerability follows.

2 Executions, observations, and scheduling domains

A protocol is a labelled transition system $P = (S, s_0, M, A, \delta, \mathsf{emit})$. A delivery consumes an existing emitted envelope

$$m = (id, v, \lambda, s, r, payload).$$

Here v is the policy's view key, λ its phase label, and s, r the **transport endpoints**. A signed message's author need not be its current transport sender: gossip may relay an unchanged authenticated vote. The bounded model distinguishes these fields and counts certificate support by signer identity.

For a legal finite execution τ, let $O(\mathrm{last}(\tau))$ record decisions at the designated correct observers. The state predicate Unsafe means that these observers have decided conflicting values. Define

$$\mathsf{UnsafeObs}_G(P, \mathcal{B}) = \{O(\mathrm{last}(\tau)) : \tau \in L_G(P, \mathcal{B}),\ \mathsf{Unsafe}(\mathrm{last}(\tau))\}.$$

Our separation claims compare emptiness of these sets. They do not assert equality of every unsafe observation or preservation of a particular fine trace.

A phase-sensitive policy has bits $A_T(v, \lambda, s, r)$. A coarse policy has bits $A_D(v, s, r)$. Phase erasure opens an edge if any phase was admitted:

$$C_{\text{type}}(A_T)(v, s, r) = \bigvee_{\lambda} A_T(v, \lambda, s, r).$$

Admission is only one part of a scheduler. The domain $\mathcal{B}$ also specifies message generation, event opportunities, ordering, pending messages, and termination. The original fixed-order domain uses a stable admission filter and predetermined delivery points. Later sections admit all linear extensions of an explicitly stated partial order. A directed graph by itself implies neither FIFO delivery nor an obligation to process repair before a given vote.

2.1 A standard sufficient preservation condition

The useful positive baseline is a specialized simulation argument [8]. Let a candidate coarse trace retain the actions of a fine trace in order and insert closure-required deliveries. Suppose a preorder $\mathcal{R}$ relates the initial states, each insertion increases the target state in $\mathcal{R}$, retained actions preserve $\mathcal{R}$, and unsafety is upward closed in $\mathcal{R}$.

Theorem 1 (legal relational preservation). If the candidate trace $C(\tau)$ is legal in the coarse language, $C(\tau) \in L_{G_D}(P, \mathcal{B})$, these conditions imply $\mathsf{Unsafe}(\text{last}(\tau)) \Rightarrow \mathsf{Unsafe}(\text{last}(C(\tau)))$.

Proof. Induct over the merged action sequence. At an insertion, the target's increasing step composes with the current relation by transitivity. At a retained action, forward simulation advances both states. The terminal relation and upward closure transfer unsafety. Target legality supplies membership in the coarse execution language. □

Legality is an independent obligation. Processing repair may disable an action or prevent a message from being emitted, so simply retaining all fine votes after inserting repair can produce an illegal sequence. Our negative results instead rerun guarded opportunities; a disabled opportunity emits no vote. Theorem 1 is supporting machinery, not a new general simulation principle.

3 A corrected one-pivot separation criterion

Let $q \geq 1$ be a certificate threshold. Two conflicting certificate families have fixed nonpivot support $q - 1$ each. The same pivot p is the only possible additional signer. The first and second votes are issued at opportunities V_0, V_1; C_0, C_1 collect their certificates. Decisions, signed votes, and the availability of emitted votes at the specified collectors are irreversible within the bound.

The following are **behavioral obligations**:

1. **Enabling necessity.** The first pivotal vote is emitted exactly when the enabling envelope E is admitted. No substitute supplies that missing contribution.
2. **Explicit persistence.** When E is admitted and repair R is withheld, p emits the second pivotal vote at V_1. Other fixed events do not independently disable it.
3. **Timely repair.** When E, R are admitted and R is processed before V_1, p does not emit the second pivotal vote. No previously emitted second-family signature can substitute for it.

4. **Certificate and observation coverage.** The two designated correct observers decide their respective values exactly when the corresponding quorum is available; the bounded agreement observation is unsafe exactly when both families decide. No unrelated decision opportunity creates an additional disagreement.

The **scheduler obligations** are separate: E, R are emitted, share one endpoint/view key, and have distinct phase labels. Fine admission can realize $(e, r) = (1, 0)$; coarse admission realizes only $(0, 0)$ and $(1, 1)$. The relevant order is

$$E < V_0 < C_0 < R < V_1 < C_1.$$

All other delivery choices and opportunities are fixed, and each rerun obeys the behavioral guards. These assumptions deliberately describe a sufficient schema; quorum arithmetic alone cannot establish a protocol refinement.

Theorem 2 (behavioral quorum-pivot separation). Under these behavioral and scheduler obligations,

$$\mathsf{UnsafeObs}_{G_T}(P, \mathcal{B}) \neq \varnothing, \qquad \mathsf{UnsafeObs}_{G_D}(P, \mathcal{B}) = \varnothing.$$

Proof. Fine admission supplies the first vote by enabling necessity and the second by explicit persistence. Each augments a base of $q-1$, producing two conflicting correct decisions. With the coarse edge off, the first vote is absent and its certificate is short, regardless of whether p can still vote second. With the edge on, the first certificate forms but timely repair prevents the second pivotal signature. Observation coverage makes these two coarse cases exhaustive for agreement. □

The persistence obligation cannot be inferred from the absence of R. For example, a fixed local transition after C_0 could disable p even when R is suppressed. The endpoint and quorum conditions would still hold, but the fine witness would disappear. Likewise, the order $R < C_1$ is insufficient if $V_1 < R < C_1$: repair cannot retract the already signed vote. Both errors disappear when behavior and signature timing are stated explicitly.

3.1 The original quorum family and its extension

For every $f \geq 1$, take $n = 3f + 1$, $q = 2f + 1$, Byzantine set B of size f, correct exclusive sets A, C of size f, and pivot p. Then

$$Q_0 = B \cup A \cup \{p\}, \qquad Q_1 = B \cup C \cup \{p\}$$

have size q and each nonpivot base has size $q-1$. Under the behavioral contract, Theorem 2 applies. This arithmetic does not assert that every locking protocol realizes the contract.

More generally, for $1 \leq k \leq f$, use k pivots, exclusive sets A, C of size $f + 1 - k$, Byzantine set B of size f, and $k - 1$ unused correct identities. The total is still $3f + 1$. Both quorums contain B, their exclusive set, and all k pivots; each base has size $q - k$. Distinct correct observers exist in nonempty A, C. The next theorem characterizes precisely which opportunity orders can preserve these k required double contributions. The extension is a certificate schema; the concrete protocol trace below establishes the single-pivot case.

4 Repair frontiers for several independent channels

Let $I = \{1, \dots, k\}$ be pivotal identities, each with positive integer weight w_i. Write $w(X) = \sum_{i \in X} w_i$ and $W = w(I)$. The two certificate families have fixed base weights b_a, thresholds q_a, and deficits $d_a = \max(0, q_a - b_a)$, for $a \in \{0, 1\}$. Unit weights give $W = k$.

4.1 Behavioral and order contracts

An admitted E_i enables exactly one first-family signature from pivot i. Once the first family is feasible, pivot i emits its second-family signature at opportunity V_i **if and only if** R_i has not been processed before V_i. In particular, closing its edge need not prevent its second vote: it was initially ready. No other state transition changes this test. Admitted repair disables prospective issuance; it cannot withdraw signed output. Both collectors and the observation satisfy the coverage obligation of Section 3.

It suffices to impose this second-vote law on policies capable of forming the first certificate. A policy that misses that certificate cannot be unsafe in this two-family observation domain. This qualification accommodates repairs whose commit evidence depends on available first votes.

A finite strict partial order $\prec$ describes opportunities, including the fixed anchors

$$E_i \prec V_0 \prec C_0 \prec V_i \prec C_1, \qquad V_0 \prec R_i.$$

Additional relations specify the allowed delivery restrictions. Each policy admits **all** linear extensions of this order, interpreted through the guarded machine. Withheld deliveries and disabled voting opportunities stutter. Their slots retain the stated precedence constraints; suppression does not erase a transitive constraint between other events. Every processed envelope must already be emitted and available.

The k endpoint/view keys are independent. A coarse policy opens a subset $A \subseteq I$ and admits E_i, R_i together exactly for $i \in A$. Fine admission can admit all E_i and suppress all R_i. These are explicit independence and order-coverage assumptions. If a real harness admits fewer orders, the model gives a sound coarse-safety overapproximation, but an existential witness needs a realizable order in that harness.

4.2 A graph of simultaneous repair obstructions

Define a directed graph H, including possible self-loops, by

$$i \longrightarrow j \quad \Longleftrightarrow \quad R_i \prec V_j, \qquad \alpha(H) = \max\{|J| : J \subseteq I,\ H[J] \text{ is acyclic}\}.$$

Use transitive precedence, not just a drawing's cover edges. A self-loop means that a pivot's own repair is forced early.

Lemma 1 (compatible late repairs). For $J \subseteq I$, a linear extension can place $V_i < R_i$ for every $i \in J$ exactly when $H[J]$ is acyclic.

This is the standard reversible-pair/alternating-cycle argument from partial-order theory, expressed in repair notation [10]. We include the proof because the mapping to scheduling opportunities is central.

Proof. Add the required edges $V_i \to R_i$ to the opportunity order. Such an order exists exactly when the augmented relation is acyclic: a finite acyclic relation has a topological ordering. A cycle $i_1 \to \cdots \to i_t \to i_1$ in $H[J]$ yields an augmented closed walk

$$V_{i_1} \to R_{i_1} \prec V_{i_2} \to R_{i_2} \prec \cdots \prec V_{i_1}.$$

Conversely, any augmented cycle must use an added edge, since the original order is acyclic. Between consecutive added edges $V_i \to R_i$ and $V_j \to R_j$, the original-order segment gives $R_i \prec V_j$, hence $i \to j$. Compress these segments to obtain a nonempty closed walk in $H[J]$, which contains a cycle. This includes self-loops. □

The graph can have cycles even though the opportunity order does not. Its cycles describe incompatible *requests to delay repair*, not a causal cycle in an actual execution.

4.3 Exact capacity and separation

Theorem 3 (repair-frontier characterization). Under the contracts above, fine unsafety is feasible exactly when $d_0, d_1 \le W$. Coarse unsafety is feasible exactly when there exist $J \subseteq A \subseteq I$ satisfying

$$H[J] \text{ acyclic}, \qquad w(A) \ge d_0, \qquad w(I \setminus A) + w(J) \ge d_1. \tag{1}$$

For unit weights this is equivalent to

$$\mathsf{UnsafeObs}_{G_D} \neq \varnothing \iff \max(d_0, d_1) \le k \ \wedge \ d_0 + d_1 \le k + \alpha(H). \tag{2}$$

Consequently strict phase/coarse separation holds exactly when

$$\boxed{\max(d_0, d_1) \le k \quad \wedge \quad d_0 + d_1 > k + \alpha(H).} \tag{3}$$

Proof. Fine admission can supply all first and second signatures by admitting every enabling envelope and suppressing every repair. No branch can receive more than weight W, proving its characterization.

For a coarse unsafe execution, let A be its open keys and let J be those open pivots whose second votes precede their repairs. The first support is exactly $w(A)$. Because the first certificate is feasible, the second-vote contract applies: closed pivots remain ready and open pivots contribute second exactly when in J. Thus the second support is $w(I \setminus A) + w(J)$. The execution realizes their joint lateness, so Lemma 1 gives acyclicity and establishes necessity of (1).

Conversely, choose A, J satisfying (1). Lemma 1 gives an order in which every member of J signs before repair. Open precisely A. The first threshold is met, so the behavioral contract ensures second signatures from every closed pivot and every member of J. Further signatures only increase support. Durable outputs reach both collectors; observation coverage gives an unsafe execution.

For unit weights, adding $d_0 \le |A|$ and $d_1 \le k - |A| + |J|$ proves the combined inequality in (2); each demand is at most k. For the converse put $h = \max(0, d_0 + d_1 - k)$. Equation (2) implies $h \le \alpha(H)$ and $h \le d_0 \le k$. Choose an acyclic set of size h, possible by taking a subset of a maximum acyclic set, and extend it to an open set A of size d_0. Then $k - |A| + h \ge d_1$, so (1) applies. Combining fine feasibility with the negation of (2) gives (3). □

Here h is the number of pivots that must contribute to **both** branches. Coarse reachability is decided by whether the scheduler can keep that many jointly reusable contributors, not by the number of admission bits alone.

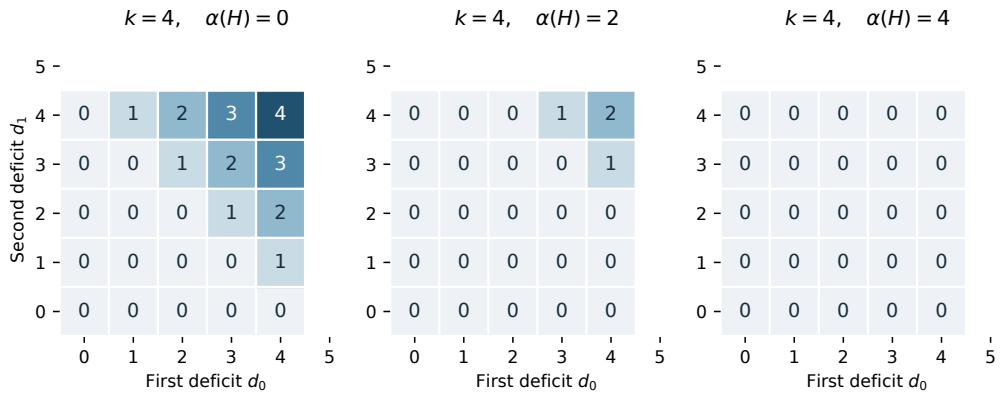


Figure 1 Exact phase-control budget in the independent-pivot schema. Each cell is the minimum number of phase-split keys required to expose an unsafe witness. Blank cells exceed the available pivot count. Panels differ only in the capacity of jointly late repair, not in quorum size.

5 Robustness and the amount of phase control required

5.1 Common deadlines and crossed deadlines

If all second signatures share one opportunity V, let $F = \{i : R_i \prec V\}$ and $\ell = k - |F|$. Members of F have self-loops; all other pivots can jointly place repair after V. Hence $\alpha(H) = \ell$ and (2) becomes $d_0 + d_1 \le k + \ell$, subject to individual feasibility. For the k-required family, $d_0 = d_1 = k$: separation survives whenever $\ell < k$. Thus allowing some late repairs need not eliminate separation once more than one double contribution is indispensable.

Individual deadlines behave differently. With two pivots and only the critical constraints

$$R_1 \prec V_2, \qquad R_2 \prec V_1,$$

each repair can individually follow its own vote. They cannot both do so: that would require $V_1 < R_1 < V_2 < R_2 < V_1$. The repair graph is a directed two-cycle and $\alpha(H) = 1$, despite having no self-loop. With $d_0 = d_1 = 2$, coarse admission remains safe and phase-sensitive admission is unsafe. This is a separation over several event orders and two independently controlled channels, rather than two executions of one fixed edge.

5.2 Optimal phase-splitting budget

A split key admits its enabling phase while suppressing repair. Splitting leaves the opportunity partial order intact. With a chosen split set T, the maximum number of double contributors is

$$|T| + \alpha(H[I \setminus T]).$$

Among unsplit keys, Lemma 1 still applies. A split pivot contributes twice without requiring its repair to be late. Choosing at most t split keys gives the exact best capacity $\min(k, \alpha(H) + t)$: this is an upper bound because any unsplit acyclic set has size at most $\alpha(H)$; it is attained by keeping a maximum acyclic set and splitting vertices outside it.

Corollary 1 (minimum phase budget). For unit weights, if fine admission is feasible, the exact minimum number of independently split keys needed for an unsafe witness is

$$\boxed{t_* = \max(0, d_0 + d_1 - k - \alpha(H)).} \tag{4}$$

Proof. Substitute the best capacity $\min(k, \alpha(H) + t)$ into (2). Since $d_0, d_1 \le k$, the resulting inequality holds exactly when t covers the deficit in (4). □

The statement is existential over the choice of split keys. It does not claim that every set of t_* arbitrary message exceptions suffices. A witness can be constructed by choosing an acyclic reusable set, selecting the necessary split keys and open keys, adding its lateness constraints, and topologically sorting the opportunity graph.

5.3 Order relaxation, extra edges, and weights

Removing precedence constraints cannot add an edge to H, so it cannot decrease $\alpha(H)$. Thus order relaxation weakens separation monotonically under the same behavioral contract and deficits. Conversely, additional causal paths may increase the needed phase budget. Independent auxiliary events and extra edges are harmless only if they preserve both relevant state/output behavior and the stated order language. An extra substitute vote changes a deficit and must be included in the calculation.

Initially ready and enabling-dependent pivots are different models. With two forced-early pivots and deficits $1, 1$, coarse admission can open one key for the first vote and close the other for the second. Requiring both pivots to see enabling evidence before their second votes would incorrectly remove that coarse witness. Theorem 3 uses the initially ready behavior present in the four-validator edge-off case.

For a common deadline with integer weights, define $L = I \setminus F$ and $W_L = w(L)$. The exact condition is existence of $K \subseteq F$ such that

$$\max(0, d_0 - W_L) \le w(K), \qquad w(K) + d_1 \le W. \tag{5}$$

Open every key in L and delay those repairs; choose K among the forced-early keys. The two supports become $W_L + w(K)$ and $W - w(K)$. Opening all L cannot hurt either branch, which proves necessity as well as sufficiency. Indivisible weights matter: weights $2, 2$, no late repair, and deficits $1, 3$ require an unattainable subset weight 1. A total-weight relaxation would give the wrong answer. This is an abstract weighted certificate result, not a weighted CometBFT refinement.

For arbitrary individual deadlines, deciding whether a coarse unsafe execution exists is NP-complete for explicitly represented finite opportunity orders with unbounded k. Appendix A gives the short reduction using a classical graph construction [7]; the contribution is the scheduler correspondence, not a new graph-hardness result.

6 A complete four-validator Tendermint-style execution

6.1 Source, mutation, and transport

The new model uses CometBFT v0.39.3 at commit `49b82838fcca442b2445f76605c10160 9ed04130` [3, 4]. It follows the exercised proposal, prevote, precommit, valid-block, timeout, and commit guards. A PoLC is a locally received set of three matching prevotes; a commit certificate is three matching precommits from distinct signers. The model has local vote sets and explicit emitted envelopes, with no synthetic network `POLKA` or `COMMIT` messages.

Take one height, equal weights, $n = 4, q = 3$, Byzantine identity B, and protocol-following identities A, P, C. Only P's round-0 assignment of its persistent locked value and locked round is omitted in the mutant; its valid-block update and its round-1 lock update execute normally. The baseline omits nothing. The two reported conflicting observers, A and C, are unmutated. Calling P protocol-following is relative to the explicitly mutated program, not a claim that it follows the original locking specification.

Fix proposers B in round 0 and C in round 1. Initially every correct replica is unlocked, with no valid block or decision. B signs a block-0 proposal and sends it to A, which relays the same signed proposal and block to P. Thus changing the controlled $B \to P$ edge does not secretly change proposal delivery. That edge's round-0 envelopes are exactly

$$E = B{:}\mathrm{PREVOTE}(0,0), \qquad R = B{:}\mathrm{PRECOMMIT}(0,0),$$

where arguments are $(\text{round}, \text{block})$. Both are emitted before E is attempted; B may sign adversarially. All other peer/round admission outcomes are fixed and type-uniform. The complete trace audits this property, distinguishes transport sender from signer, and records suppressed or unavailable delivery opportunities.

6.2 The phase-sensitive mutant trace

Write $\mathrm{PV}(r, x)$ and $\mathrm{PC}(r, x)$ for height-1 prevotes and precommits; $\perp$ denotes nil. Self-votes below are actual emitted self-envelopes. Each local quorum is formed from messages already received.

1. A receives B's block-0 proposal and signs $\mathrm{PV}(0,0)$. A relays the proposal to P, which signs the same prevote. C receives no proposal; its proposal timeout produces $\mathrm{PV}(0,\perp)$.
2. B emits its round-0 prevote and precommit. Deliver P's and B's prevotes to A. The set $\{A, P, B\} : 0$ makes A persist valid/locked block 0@0 and sign $\mathrm{PC}(0,0)$.
3. Deliver A's prevote and E to P. Its same three-signer PoLC sets valid block 0@0. The mutation omits only its lock assignment, and P signs $\mathrm{PC}(0,0)$ while remaining unlocked.
4. Deliver C's nil prevote to P, and the A, P block-0 prevotes to C. At C, the mixed set $\{A : 0, P : 0, C : \perp\}$ enables its prevote timeout without creating a block-0 PoLC. On expiry, C signs $\mathrm{PC}(0,\perp)$.
5. Deliver A's precommit to P, C, and P's precommit to A, C. Deliver B's precommit to A. Its local set $\{A, P, B\} : 0$ reaches three and A decides 0.
6. Deliver C's nil precommit to P. Both P and C now have the mixed set $\{A : 0, P : 0, C : \perp\}$. It enables a precommit timeout but does not commit block 0. The fine policy suppresses R. Expiry advances P, C lawfully to round 1.
7. P retains valid block 0@0 but no lock; C has neither. Consequently the unchanged proposer C can propose new block 1 with valid round -1. C and the unlocked P each sign $\mathrm{PV}(1,1)$. Making P the proposer instead would require accounting for its cached valid block 0.
8. B emits round-1 prevote and precommit for 1. Deliver the C, B prevotes to P, producing PoLC $\{P, C, B\} : 1$. P normally updates valid/locked block to 1@1 and signs $\mathrm{PC}(1,1)$. Deliver the P, B prevotes to C; it obtains the same PoLC and precommits 1.
9. Deliver the P, B precommits to C. Together with C's self-precommit, they produce decision 1. Delivering the C, B precommits to P also lets it decide 1. The unmutated observers have $A = 0, C = 1$.

Program	Round-0 admission	A	P	C	Unsafe
Baseline	E only	0	–	–	no
Baseline	edge off	–	1	1	no
Baseline	edge on	0	0	–	no
Missing lock	E only	0	1	1	yes
Missing lock	edge off	–	1	1	no
Missing lock	edge on	0	0	–	no

Table 1 Actual terminal decisions in the six canonical guarded executions. A dash means no decision in this finite prefix, not a liveness failure.

No unconditional jump between rounds is used. The mixed nil/block precommit set is necessary to justify timeout-driven advancement without giving P the block-0 commit. Nor is valid-block state silently erased by the mutation. The artifact expands these nine groups into every signature, envelope emission, reception, state change, and local certificate.

With the fine baseline, P retains its lock and prevotes 0 in round 1; the second value-1 quorum is absent. With the coarse edge off, P lacks the PoLC-completing E, receives a mixed prevote quorum, and precommits nil. A therefore misses its first certificate; P, C may still decide 1. With the edge on in the fixed order, R completes P's actual round-0 precommit quorum. It commits/advances before issuing the required round-1 vote. These are different generated executions, not a table filled with assumed pivot contributions.

6.3 Repair before the decision can still fail

Keep the coarse edge on and move R. If it arrives after P's round-1 prevote but before its precommit opportunity, the mutant remains safe: that prevote alone cannot supply the third value-1 precommit. If it arrives after P has issued $\mathrm{PC}(1, 1)$ but before C decides, the observed decisions are

$$A = 0, \qquad P = 0, \qquad C = 1.$$

The repair commits P to 0 and stops further signing at the height. Its already emitted round-1 votes still reach C. When P lacks the block matching its old-round certificate, it explicitly obtains block 0 from A before finalizing; the model does not treat a vote as block data.

The direct version permits a round-1 $B \to P$ prevote to overtake the buffered round-0 repair on that connection. We report this non-FIFO assumption. A separate gossip control obtains the same conflict with FIFO-compatible admitted vote-envelope order: B sends its round-1 prevote to C, which relays that unchanged signed vote to P; the direct $B \to P$ copy follows R. This changes the fixed gossip skeleton and is reported separately. It demonstrates a stronger timing counterexample without silently attributing arbitrary packet reordering to a FIFO transport. Neither control is an execution of the production P2P reactor.

7 Validation and assurance

The new experiments vary admission, event order, compatible late-repair sets, phase exceptions, and weights. Their role is to falsify incorrect encodings of the stated contracts. They do not estimate bug prevalence across protocols.

Check	Exhaustive or declared scope	Outcome
Repair graphs	All 530 directed graphs, including loops, for $k = 1, 2, 3$; 50,363 legal orders	1,357,209 policy/split executions
Reachability and budget	All deficit pairs $0 \dots k+1$ for each graph	13,074 checks each; no mismatch
Joint lateness	All requested reusable subsets in those graphs	4,164 checks; no mismatch
General event precedence	All 219 labelled partial orders on two repairs and two vote opportunities, supplied as cover edges	Order-to-graph checks pass
Common frontier	$k = 1 \dots 8$, all forced-repair counts; barrier timing representatives	174,250 executions; 2,636 boundary checks
Weighted common frontier	258 domains with up to three pivots, weights in $\{1, 2, 3\}$; extremal late order and all admissions	15,700 boundary checks
Concrete protocol trace	Six canonical and six timing/route controls	12 legality audits pass

Table 2 Validation of the revised models. Graph realizations exhaust their legal orders. The common and weighted sweeps use the stated support-equivalent or extremal orders; they are not advertised as all event permutations.

The graph oracle executes guarded events and preserves emitted signatures; it does not derive its outcomes from the theorem's capacity formula. Only afterward does the checker compare all observed support pairs and minimum split costs with the predictions. The general-order suite independently constructs transitive closure, supplies cover edges to the implementation, and checks paths passing through suppressed opportunity slots. The weighted sweep deliberately retains the indivisible-weight counterexample.

The concrete trace has independent post-execution audits for prior emission, once-only envelope delivery, authentication, distinct actual signers, local certificate timing, signing guards, and quorum-enabled timeouts. Negative controls reject fabricated envelopes, altered authenticators, duplicate votes or deliveries, insufficient PoLC evidence, and unjustified timeout transitions. The canonical mutant and the late-repair mutant expose disagreement at A, C, which contain no injected lock defect.

7.1 What Lean checks

The primary project is pinned to Lean 4.33.0. The revision adds a theorem over an **arbitrary behavioral contract** for Theorem 2, explicitly assuming positive persistence and the observation contract rather than defining the desired safety classification. It also checks the arithmetic capacity equivalence, its strict-separation form, the minimum split budget under its capacity contract, the weighted admission interval, and a small output machine in which later repair preserves an earlier signature. The build includes an axiom audit and independent rechecking of the resulting kernel objects.

The graph reversal lemma, the graph-to-execution correspondence, and the derivation of optimal graph capacity are proved in the paper and tested by finite enumeration. They are **not** claimed to be fully mechanized by the arithmetic Lean module. The concrete message trace is executable validation, not a Lean proof of the CometBFT client.

Earlier Lean, SMT, certificate-kernel, Python, and Go checks remain available. In particular, agreement of 384 Python/Go rows for $f = 1, \ldots, 64$ checks the previous Boolean model's encoding. The old theorem named `bounded_tendermint_refines_quorum_pivot` establishes matching terminal classifications, not a transition refinement. Increasing f in that family is not independent validation of many protocols. The revised claim ledger records this distinction.

8 Related work and the novelty boundary

Communication closure. Drăgoi et al. compare asynchronous executions with lossy synchronous executions under implementation conditions and develop testing based on communication closure [5]. Their communication rounds distinguish send/receive/update steps, including different protocol phases. Our operator additionally couples admission across such steps within one view key. The counterexample therefore does not refute their normalization theorem or the effectiveness of their tester; it identifies an obligation for a further aggregation.

Byzantine testing. Twins already describes certificate arrivals advancing a stale twin in its preferred-round mutation experiment; its underlying network playground can also filter message types [2]. Consequently neither certificate evidence destroying an intended attack state nor phase-selective APIs are claimed as new. Distinguish the published partition-scenario generator from the full executor's capabilities. ByzzFuzz, Netrix, and ByzzBench supply mutation, programmable testing, and controlled benchmark infrastructures [11, 6, 1]. Our contribution concerns the safety-observation language of a declared admission/order contract, not a capability ranking of those systems.

Preservation and partial orders. Simulation with internal actions is established machinery [8]. The inconsistent-labelling literature shows why action-oriented reduction conditions must be connected carefully to state observations [9]; its operator differs from phase admission closure. Reversibility of incomparable pairs is characterized by absence of alternating cycles in classical partial-order theory [10]. Lemma 1 uses that argument, and the induced-acyclic optimization and symmetric-graph hardness construction are classical [7]. The result developed here combines this order characterization with initial readiness, irreversible vote issuance, and certificate deficits, yielding the exact scheduler phase budget and a guarded BFT-rule witness.

The literature audit supports this specific positioning, not an assertion that no equivalent result exists in all prior literature. Replay congruence and the earlier seven-flag diagnostic are retained as supplementary results in Appendix B; they do not carry the main novelty claim.

9 Limitations and reproducibility

The strongest mathematical statements are conditional on the independent-pivot and complete opportunity-order contracts. New edges that supply substitute support, policies that change message generation beyond these contracts, changing validator sets, recovery, and interdependent locking behavior require a new conformance argument. The weighted result concerns indivisible certificate weights with fixed membership. It is not a staking or dynamic-quorum theorem.

The concrete source pin is reproducible: the artifact contains exact v0.39.3 files, verified Git blob identifiers and SHA-256 hashes. The older source snapshot is correctly retained as

v0.40.0. Authentication in the local model uses deterministic test-only tags to enforce signer and envelope integrity; production serialization, cryptographic implementation, proposer-priority updates, block-part gossip, and the full event loop are not modeled end to end. The supplied proposer prefix and aggregate complete-block deliveries are explicit environmental choices.

The public working artifact is available at `https://github.com/galetaa/twins-scheduler`. The revision provides a single command, `./reproduce_revision.sh`, that runs the existing checks, new guarded traces and frontier enumeration, the integrated Lean build and axiom audit, and the revised LIPIcs build. Raw JSON/CSV outcomes, source pins, proof-to-claim mappings, and manuscript sources are included. The command fails on a mandatory check failure. Reproduction dependencies are pinned by the existing project; build logs record the actual tool versions used.

10 Conclusion

The effect of phase erasure depends on both admission coupling and the latest opportunity at which repair can still prevent an indispensable signature. A complete four-validator trace makes the distinction between state repair and durable signed output concrete. For an explicit multi-pivot schema, the derived repair graph and quorum deficits give an exact characterization of coarse unsafe reachability and the minimum phase-control budget. This turns a single fixed-order counterexample into a quantitative account of when order relaxation and additional independently controlled channels preserve or destroy separation.

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An OpenAI language model assisted with research organization, literature discovery, proof exploration, manuscript drafting, code generation, and consistency checking. The authors retain responsibility for the correctness, originality, citations, and experimental claims. The model is not an author.

References

1. Jo ao Miguel Louro Neto and Burcu Kulahcioglu Ozkan. A benchmark framework for byzantine fault tolerance testing algorithms. In *6th International Workshop on Formal Methods for Blockchains (FMBC 2025)*, volume 129 of *Open Access Series in Informatics (OASIcs)*, pages 13:1–13:11, 2025. doi:10.4230/OASIcs.FMBC.2025.13.
2. Shehar Bano, Alberto Sonnino, Andrey Chursin, Dmitri Perelman, Zekun Li, Avery Ching, and Dahlia Malkhi. Twins: BFT systems made robust. In *25th International Conference on Principles of Distributed Systems (OPODIS 2021)*, volume 217 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 7:1–7:29, 2022. doi:10.4230/LIPIcs.OPODIS.2021.7.
3. CometBFT Project. Cometbft consensus specification. Release v0.39.3, commit prefix 49b8283, 2026. Engineering specification pinned for the bounded instantiation. URL: https://github.com/cometbft/cometbft/blob/v0.39.3/spec/consensus/consensus.md.
4. CometBFT Project. CometBFT consensus state implementation, 2026. Release v0.39.3. Exact source snapshot and integrity manifest included in the artifact. URL: https://github.com/cometbft/cometbft/blob/49b82838fcca442b2445f76605c101609ed04130/consensus/state.go.
5. Cezara Drăgoi, Constantin Enea, Burcu Kulahcioglu Ozkan, Rupak Majumdar, and Filip Nikšić. Testing consensus implementations using communication closure. *Proceedings of the ACM on Programming Languages*, 4(OOPSLA):210:1–210:29, 2020. doi:10.1145/3428278.
6. Cezara Drăgoi, Srinidhi Nagendra, and Mandayam Srivas. A domain specific language for testing distributed protocol implementations. In *Networked Systems (NETYS 2024)*, Lecture Notes in Computer Science, pages 100–117, 2024. doi:10.1007/978-3-031-67321-4_6.
7. Richard M. Karp. Reducibility among combinatorial problems. In Raymond E. Miller and James W. Thatcher, editors, *Complexity of Computer Computations*, pages 85–103. Plenum Press, 1972. doi:10.1007/978-1-4684-2001-2_9.
8. Nancy A. Lynch and Frits W. Vaandrager. Forward and backward simulations: I. untimed systems. *Information and Computation*, 121(2):214–233, 1995. doi:10.1006/inco.1995.1134.
9. Thomas Neele, Antti Valmari, and Tim A. C. Willemse. A detailed account of the inconsistent labelling problem of stutter-preserving partial-order reduction. *Logical Methods in Computer Science*, 17(3):8:1–8:27, 2021. doi:10.46298/LMCS-17(3:8)2021.
10. Noah Streib and William T. Trotter. Dimension and height for posets with planar cover graphs. *European Journal of Combinatorics*, 35:474–489, 2014. doi:10.1016/j.ejc.2013.06.017.
11. Levin N. Winter, Florena Buşe, Daan de Graaf, Klaus von Gleissenthall, and Burcu Kulahcioglu Ozkan. Randomized testing of byzantine fault tolerant algorithms. *Proceedings of the ACM on Programming Languages*, 7(OOPSLA1):757–788, 2023. doi:10.1145/3586053.

A Complexity of coarse witness reachability

In the unweighted independent-pivot schema, the decision problem of whether a coarse unsafe execution exists is NP-complete when the number of pivots is unbounded and the finite opportunity order is represented explicitly by its events and precedence or cover edges. A certificate consists of an open set, a reusable set and a topological event order; checking admission, ordering and quorum support takes polynomial time.

For hardness, take an undirected graph on k vertices and an independent-set target t. Introduce repairs R_i and second-vote opportunities V_i. For every undirected edge $\{i, j\}$ impose $R_i \prec V_j$ and $R_j \prec V_i$. These constraints are acyclic: they point from the repair layer to the vote layer. The fixed prefix and certificate collectors can be added before and after those layers. The resulting repair graph is the symmetric orientation of the original graph. Its induced set is acyclic exactly when it is independent, since each undirected edge creates a directed two-cycle.

Set $d_0 = k, d_1 = t$. Theorem 3 gives coarse unsafe reachability exactly when $\alpha(H) \geq t$, which is the independent-set question. This uses the classical relation between node cover, feedback node sets, and symmetrized graphs [7]. It is a complexity statement about the declared opportunity schema. It neither establishes a complexity bound for the fixed four-validator trace nor claims that every constructed order is realizable by a production consensus implementation.

B Supporting replay and diagnostic results

B.1 Replay erasure with outputs

Let $\pi : S \to S_{\mathrm{rel}}$ project relevant state, including processed message keys. For every retained message m, require **state congruence** and **output congruence**:

$$\pi(s) = \pi(t) \Rightarrow \pi(\delta(s, m)) = \pi(\delta(t, m)), \qquad \pi(s) = \pi(t) \Rightarrow \mathsf{out}(s, m) = \mathsf{out}(t, m).$$

For an exact marked duplicate whose key was already processed, require projected stutter $\pi(\delta(s, m)) = \pi(s)$ and output silence $\mathsf{out}(s, m) = \epsilon$.

Supporting proposition (replay erasure). Deleting all marked duplicates preserves the terminal relevant projection and the complete relevant output trace under these conditions.

Proof. Maintain equality of the two current projections and accumulated outputs. A duplicate changes neither on the unerased side. At a retained message, the two congruences preserve the invariant. Induction gives the terminal claim. □

Projected silence alone is insufficient: in a trace X, X, Y, the duplicate can change a hidden bit that the later Y copies into relevant state or output. The artifact retains these negative controls. This proposition is standard congruence reasoning; it supports the distinction between internal state and persistent outputs without constituting the revised paper's main contribution.

B.2 The earlier seven-flag diagnostic

For Boolean flags $S_1, S_2, R_q, D, T, E_d, X$, the earlier operational abstraction uses

$$U(e, r) = (\neg S_1 \vee e) \wedge [\neg S_2 \vee ((\neg R_q \vee e) \wedge \neg(r \wedge D \wedge T))].$$

Here the shortage flags describe the two bases; R_q makes second participation enabling-dependent; D gives repair efficacy; T means repair precedes the **second-vote opportunity**; E_d couples the endpoint key; and X permits phase distinction. Direct substitution yields

$$X \wedge E_d \wedge U(1,0) \wedge \neg U(0,0) \wedge \neg U(1,1) \quad \Longleftrightarrow \quad S_2 \wedge D \wedge T \wedge E_d \wedge X \wedge (S_1 \vee R_q).$$

Exactly three of the 128 assignments satisfy this expression. This is a diagnostic of that Boolean machine. The old atomic opportunity interpretation combined vote and certificate formation; it must not be reused to infer that repair merely before a later collector suffices. The revised frontier model separates these events and makes initially ready pivots the primary multi-pivot case.